BST Physics Curriculum Vol 10 Chapter 1 — Linear Algebra + Hilbert Spaces v0.4 (Saturday Wave 2 reader-grade polish; Cal cold-read ready)

Lyra (Claude 4.7) [Vol 10 primary]

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Vol 10 Chapter 1 — Linear Algebra + Hilbert Spaces

Chapter motivation

Standard mathematical-methods textbook treatment of linear algebra + Hilbert spaces (finite-dim then L^2). BST cross-reference: Bergman H^2(D_IV^5) is the substrate's canonical Hilbert space (Vol 1 Ch 2 + T2428); standard $L^2(\mathbb{R}^3)$ emerges as macroscopic limit (Vol 5 Ch 1 pedagogical bridge).

Section 1.0b — Reader-grade pedagogy

Level 1: Standard linear algebra + Hilbert space machinery; BST cross-link: Bergman H^2(D_IV^5) is the substrate canonical Hilbert space; $L^2(\mathbb{R}^3)$ emerges as macroscopic limit.

Level 2 (graduate): Inner product spaces + Cauchy completeness + L^2-spaces + bounded operators + spectral theorem + self-adjoint operators (standard methods); BST cross-link: Bergman H^2 as concrete substrate-Hilbert-space example with reproducing kernel K_B(z, w̄) = c_FK · h(z, w̄)^(−g/rank); c_FK · π^((g+rank)/rank) = 225 EXACT per T2442.

Level 3 (5th-grader): Hilbert spaces are mathematical “rooms” where quantum-state vectors live. Standard math methods cover all the machinery (bra-ket, spectral theorem, completeness). BST shows the substrate has a specific Hilbert space (Bergman) — standard $L^2(\mathbb{R}^3)$ emerges as the large-scale limit.

— Lyra, Vol 10 Ch 1 v0.3, Saturday 2026-05-23 EDT